MPU9250 -
we are communicating using SPI interface

SPI mode for MPU9250
supports SPI mode 0 [CPOL=0 & CPHA=0]
CPOL=0 → when clock is idle
CPHA=0 → Data is sampled on rising edge of SCLK & shifted out on the falling edge

- SPI works in full-duplex mode.

SPI signals.
SCLK → serial clk [driven by FGPA]
MOSI → master out slave in [FGPA → MPU9250]
MISO → master in slave out [MPU9250 → FGPA]
CS → chip select [Active low]
↳ To start communication

MSB is transmitted first.

SPI Read operation.
1. set CS=0 [start communication]
2. send the reg address with MSB=1 [read operation]
3. send dummy byte [0X00], while receive reg data.
4. set CS=1 [end commun]

Q:	MOSI [FG→MPU]	MISO [MPU - FGPA]
1.	0X75	0X80 (Read reg)
2.	0X00	0X71 [Response]

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SPI ~~read~~ write operation

1. Set CS = 0
2. Send the Register address with MSB = 0 (W.R.O.M)
3. Send the data byte to be written
4. Set CS = 1.

Ex: writing to PWR_MGMT_1 [0x6B].

	MOSI [FPGA → MPU]	MISO (MPU → FPGA)
1.	0x6B	0x00 [would rest]
2.	0x01 [would run]	

I/P	O/P
clk	mosi
reset	sclk
[7:0] tx-data	cs
miso	[7:0] rx-data
	busy

FPGA to sensor

1 miso-data = 0x3C

= 00111100

sending data ← tx-data = 0x75

• slave sends 0x3C via MISO

at each clk it will sends bit

miso = miso-data[bit index]

master sends → 0x75 [01110101] (MOSI)

slaves recv → 0x3C [00111100] (MISO)

output seen → MISO [sensor output]

given input → tx-data